

Connection between CCA and Contrastive Learning

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1 Setup

In this note, we show a connection between the solution to Canonical Correlation Analysis (CCA) and Contrastive Learning with linear encoders. Let $u \in \mathcal{U}$ and $v \in \mathcal{V}$ be random variables corresponding to two data modalities, possibly in different spaces (not necessarily Euclidean or of the same dimensions). Our goal is to find encoders mapping into a common low-dimensional Euclidean embedding space:

$$\begin{aligned} g_u : \mathcal{U} &\rightarrow \mathbb{R}^{d_e} \\ g_v : \mathcal{V} &\rightarrow \mathbb{R}^{d_e}, \end{aligned}$$

so that $g_u(u)$ and $g_v(v)$ are aligned if u, v are two representations of the same underlying object, e.g., an image and corresponding caption, and $g_u(u)$ and $g_v(v)$ are misaligned otherwise.

Data assumption: We collect N pairs of i.i.d. samples $\{(u_i, v_i)\}_{i=1}^N \sim \mu$ drawn from the joint distribution $\mu \in \mathcal{P}(\mathcal{U} \times \mathcal{V})$.

For the remainder of this note, we will assume the u, v live in Euclidean spaces (possibly high-dimensional for text and images), i.e., $\mathcal{U} = \mathbb{R}^{d_u}$ and $\mathcal{V} = \mathbb{R}^{d_v}$ and the encoders are linear:

$$\begin{aligned} g_u(u) &= Gu, \quad G \in \mathbb{R}^{d_e \times d_u} \\ g_v(v) &= Hv, \quad H \in \mathbb{R}^{d_e \times d_v}. \end{aligned}$$

2 Contrastive Learning

The population-level loss function in contrastive learning is given by

$$\mathcal{L}(G, H) := \mathbb{E}_{(u,v) \sim \mu} \langle Gu, Hv \rangle - \frac{1}{2} \mathbb{E}_{u \sim \mu_u} \log(\mathbb{E}_{v \sim \mu_v} \exp(\langle Gu, Hv \rangle)) - \frac{1}{2} \mathbb{E}_{v \sim \mu_v} \log(\mathbb{E}_{u \sim \mu_u} \exp(\langle Gu, Hv \rangle)).$$

Note that given a finite number of samples, the corresponding empirical loss has the common form:

$$\mathcal{L}^N(G, H) := \sum_{i=1}^N \langle Gu_i, Hv_i \rangle - \sum_{i=1}^N \log \left(\sum_{j=1}^N \exp(\langle Gu_i, Hv_j \rangle) \right) - \sum_{i=1}^N \log \left(\sum_{j=1}^N \exp(\langle Gu_j, Hv_i \rangle) \right).$$

If μ is multivariate Gaussian, we can compute the expectations in $\mathcal{L}$ explicitly; see Lemma E.2 in [1]. The resulting loss function is equivalently given by

$$\mathcal{L}(G, H) = \text{Tr}(G^T H \Sigma_{vu}) - \frac{1}{2} \text{Tr}(H^T G \Sigma_{uu} G^T H \Sigma_{vv}).$$

We note that this objective only depends on the matrix product $A := G^T H$. Then, minimizing $L(A) = \text{Tr}(A \Sigma_{vu}) - \frac{1}{2} \text{Tr}(A^T \Sigma_{uu} A \Sigma_{vv})$ over rank d_e matrices for A , Theorem 5.1 in [1] shows that the solution is given by an SVD

$$U \Sigma V^T = \Sigma_{uu}^{-1/2} \Sigma_{u,v} \Sigma_{vv}^{-1/2}, \quad A^* = \Sigma_{uu}^{-1/2} U_{:d_e} \Sigma_{:d_e} V_{:d_e}^T \Sigma_{vv}^{-1/2},$$

where the notation denotes that we take the first d_e columns of the matrices U and V .

3 Canonical Correlation Analysis

CCA aims to solve the optimization problem:

$$\max_{G, H} \frac{\sum_{i=1}^N u_i^T G^T H v_i}{\left(\sum_{i=1}^N u_i G^T G u_i\right) \left(\sum_{i=1}^N v_i H^T H v_i\right)},$$

which is equivalent to solving the unconstrained problem

$$\max_{G, H} \sum_{i=1}^N u_i^T G^T H v_i \quad \text{s.t.} \quad \sum_{i=1}^N u_i^T G^T G u_i = 1, \sum_{i=1}^N v_i H^T H v_i = 1.$$

Letting $\hat{\Sigma}_{u,v}$ denote empirical cross-covariance of (u, v) and $\hat{\Sigma}_u$ and $\hat{\Sigma}_v$ denote marginal covariances, we can write this problem as:

$$\max_{G, H} \text{Tr}(H \hat{\Sigma}_{uv} G^T) \quad \text{s.t.} \quad G \hat{\Sigma}_{uu} G^T = I_{d_e}, H \hat{\Sigma}_{vv} H^T = I_{d_e}.$$

After a change of variables $\tilde{G} = G \Sigma_{uu}^{1/2}$ and $\tilde{H} = H \Sigma_{vv}^{1/2}$, we have

$$\max_{\tilde{G}, \tilde{H}} \text{Tr}(\tilde{H} \hat{\Sigma}_{uu}^{-1/2} \hat{\Sigma}_{uv} \hat{\Sigma}_{vv}^{-1/2} \tilde{G}^T) \quad \text{s.t.} \quad \tilde{G} \tilde{G}^T = I_{d_e}, \tilde{H} \tilde{H}^T = I_{d_e}.$$

The solution to this problem is then given by an SVD:

$$U \Sigma V^T = \hat{\Sigma}_{uu}^{-1/2} \hat{\Sigma}_{uv} \hat{\Sigma}_{vv}^{-1/2}, \quad \tilde{H} = U_{:d_e}, \tilde{G} = V_{:d_e}.$$

Performing the transformation back to G and H gives us the same solution as for contrastive learning up to some scaling of the eigenvalues.

References

- [1] Ricardo Baptista, Andrew M Stuart, and Son Tran. A mathematical perspective on contrastive learning. *arXiv preprint arXiv:2505.24134*, 2025.